

Interfaces for CAD/CAM

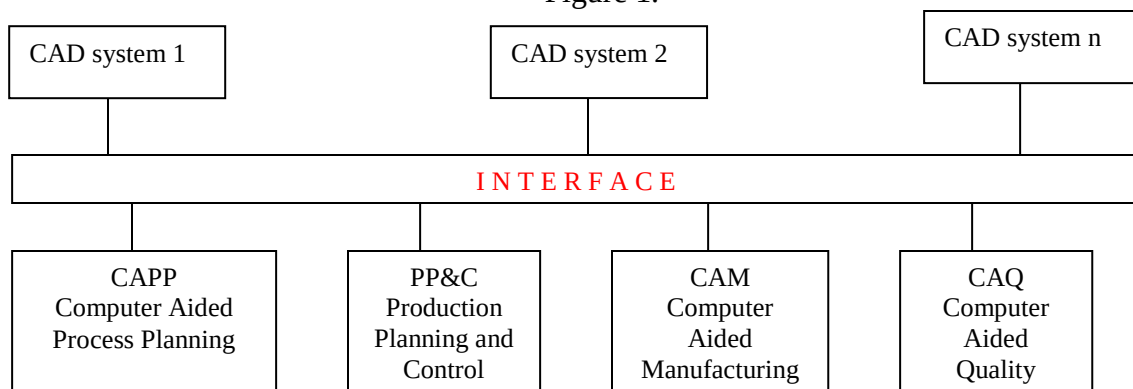
Since the introduction of the computer as an automation tool for planning and control of manufacturing operations, the problem of interconnecting various software and hardware systems has become a major issue. The first computer-based planning and control systems were custom designed and configured for dedicated applications. It was extremely difficult and often impossible to take existing software and hardware modules developed for one plant and use them for the configuration of a planning and control system for another plant. This problem had been recognized many years ago and initiated standardization activities in almost every industrialized country. Also an attempt was made to draft standards on an international level to conceive standard interfaces for interconnecting CAD and CAM systems in order to transfer design data to the planning activities of manufacturing. Information which is communicated across the interfaces contains graphical, drawing related, geometry and product model data. This data is used for process planning, scheduling, manufacturing and quality control.

The concept of a universal data exchange system is depicted in Figure 1. In this figure it is shown how various CAD systems communicate with the production planning and control system. The adaptation of protocols, data formats and data transmission rates is done by the interface. The interface must provide electrical and physical compatibility and it must ensure that the semantics of the exchanged information is maintained. Data conversion may be done directly in the interface with preprocessors, or postprocessors. With the help of such an interface it becomes possible to use existing software and hardware modules for various applications in a factory. In addition, a company's own manufacturing data can be easily merged with those from vendors and suppliers. It is also possible to build manufacturing planning and control systems from heterogeneous modules supplied by different vendors.

There are certain requirements which a standardized interface must be capable of providing. They are:

- The interface must be capable of handling all manufacturing data.
- There should be no information loss when data is transferred between heterogeneous systems. In other words, it must be possible to maintain the semantics during conversion.
- The system must be efficient to be capable of handling the realtime requirements of a manufacturing system.
- The system should be open-ended to permit extensions or contractions.
- The system should be adaptable to other standards.
- The system must be independent of the computer and communication architecture used.

Figure 1.



- The number of data entities to be handled should be a minimum.
- It must be possible to form application-oriented subsets of the standard to reduce cost and overhead.
- A logic decoupling of data structure and physical memory format is necessary to ensure a separation of the specific information structure and the transmitted data units.
- The system should be capable of achieving production data.
- The interface must be upward- and downward-compatible in a hierarchical control structure.
- Test procedures must be provided to verify the efficiency and accuracy of data transmission.

Types of Interfaces

An interface can be regarded as an aggregate of conditions, rules and conventions which describes the information exchange, between two communicating objects. An object in our context can be human, software, modules, computer hardware or manufacturing processes. In principle, three types of interfaces can be distinguished including language interfaces, procedural interfaces and descriptive interfaces. The CAD system, must be capable of exchanging data with the manufacturing world, and the interfaces are the windows to this world.

The **language** or user interface is the window through which the user can communicate with the CAD module. The user employs the icons, pictorials, menus or textual languages to describe design and manufacturing problems. The textual languages are still the most important means of user/system communication. However, the interactive graphic languages are increasingly gaining importance, they are easier to learn and offer a more natural method of user/agent communication.

The **procedural** interfaces refer to language interfaces which are necessary to do the technical calculations, CAD modeling, process planning and machine programming. These interfaces may use subprograms, parameter descriptions or host languages to perform the individual CAD activities; thereby the host language concept is of particular importance. A host language is an existing language such as Fortran, C, ... which has been extended to be able to handle specific constructs like the description of a circle. This is often done via subroutine calls. Procedural interfaces are also used to set up, in a generic way, the CAD configuration to be used or to manipulate data of-the CAD-database (ex GKS).

Descriptive interfaces are the windows to other CAD systems, to the master databases or to the manufacturing activities. With this type of interfaces it is possible to exchange models between two different CAD systems. For example, model data of the CAD system A is converted via a pre-processor to a neutral presentation form and from there via post-processor to the representation scheme needed by the CAD system B. The exchange model representation is a system-independent method of describing the application. It can be accessed by any CAD system which has the proper pre- and post-processors. Typical equipment independent interfaces to manufacturing are CLDATA for NC machine programming and IRDATA for robot programming.

The following interfaces are more used:

1. IGES (Initial Graphics Exchange Specification)
2. PDDI (Product Definition Data Interface) – Air Force, SUA
3. PDES (Product Data Exchange Specification) (IGES+PDDI)
4. SET (Standard d'Echange et de Transfert) – France
5. VDAFS (Verband der Automobilindustrie Flächenschnittstelle) – Association of the German Automobile Industry
6. STEP (Standard for External Representation of Product Data) – ISO WG184 – uses the EXPRESS language